Check for updates

THYROID
Volume 35, Number 11, 2025
© Mary Ann Liebert, Inc.
DOI: 10.1177/10507256251390877

SPECIAL ARTICLE

Executive Summary of the 2025 American Thyroid Association Management Guidelines for Adult Patients with Differentiated Thyroid Cancer

Stephanie Smooke Praw,[1] Benjamin J. Gigliotti,[2] Alex Tessnow,[3] Hyunseok Kang,[4] and Debra J. Margulies[5]

Background: Developed by members of the American Thyroid Association (ATA) Clinical Affairs Committee, this executive summary of the 2025 ATA guidelines for adult patients with differentiated thyroid cancer provides a summary of key points and recommendations with an emphasis on notable differences between the 2025 and 2015 guidelines.
Summary: The updated guidelines emphasize individualized care through the DATA framework (Diagnosis, risk/benefit Assessment, Treatment decisions, and response Assessment) with the goal of enhancing shared decision-making and personalized care. Highlights include expanded role of molecular diagnostics, refined risk stratification, greater emphasis on active surveillance and lobectomy, inclusion of ablative procedures, and selective use of external beam radiation therapy and chemoradiotherapy. De-escalation of surveillance for low-risk patients and introduction of the concept of complete remission are also new.
Conclusions: This executive summary aims to provide a summary of key points and recommendations with an emphasis on notable differences between the 2025 and 2015 guidelines.

Keywords: American Thyroid Association (ATA) guidelines, differentiated thyroid cancer, pathology-thyroid, surgery-adult thyroid

Introduction and Rationale

Since the publication of the 2015 American Thyroid Association (ATA) guidelines, there have been changes in the epidemiology, classification, and management of differentiated thyroid cancer (DTC).[1] The updated 2025 guidelines are informed by a decrease in the incidence of DTC, insights into thyroid cancer genomics, expanded molecular diagnostics, enhanced management options for advanced thyroid cancer, and increased emphasis on shared decision-making with the goal of creating a personalized treatment approach.[2]

The 2025 guidelines employ a new approach to clinical decision-making, referred to as DATA-**D**iagnosis, risk/benefit **A**ssessment, **T**reatment decisions, and response **A**ssessment. Interventions are assessed using individual patient factors to determine whether a particular intervention is appropriate. This approach facilitates shared decision-making when multiple management strategies are available. Following intervention, the 2025 ATA risk assessment tool should be used to evaluate response and guide decisions regarding ongoing monitoring and/or additional treatment, from initial diagnosis through survivorship.

The 2025 guidelines also include several Good Practice Statements as alternatives to graded recommendations when the quality of evidence is low or insufficient. These are intended to be strong recommendations applicable to nearly all persons or situations and required unanimous consensus from the guideline group.

The Clinical Affairs Committee of the ATA has developed this executive summary of the 2025 ATA guidelines on DTC to provide a summary of key points and recommendations with an emphasis on notable differences between the 2025 and 2015 guidelines. Recommendation (R) numbers, figures, and tables referenced throughout correlate directly with the 2025 ATA guideline document.

[1]Division of Endocrinology, Diabetes, and Metabolism, Department of Medicine, University of California Los Angeles David Geffen School of Medicine, Los Angeles, California, USA.
[2]Department of Medicine, University of Rochester School of Medicine and Dentistry, Rochester, New York, USA.
[3]Division of Endocrinology, Department of Internal Medicine, Southwestern Medical Center, University of Texas, Dallas, Texas, USA.
[4]Department of Medicine, Stanford University, Palo Alto, California, USA.
[5]Division of Endocrinology, Summit Health, Morristown, New Jersey, USA.
This Executive Summary was completed at the request of the ATA and the 2025 Differentiated Thyroid Cancer Guidelines Committee. Permission to reproduce figures and tables was given by the ATA. The complete reference list can be found in the full guidelines.

Key New Recommendations/Key Changes

Initial management of DTC

There are several new and updated recommendations regarding the preoperative management of DTC (Table 1). It is now explicitly recommended that thyroid cancer surgery be performed by a high-volume surgeon (≥25 thyroidectomies/year) (R6). Preoperative neck ultrasound to evaluate for the presence of invasion of tumor into surrounding structures and for cervical lymphadenopathy is also strongly recommended (R7). Fine needle aspiration with thyroglobulin (Tg) needle washout is again advised for lymph nodes with a short axis of ≥8 mm, if this would impact surgical management (R7). CT or MRI of the neck and mediastinum with intravenous contrast is recommended if there is bulky lymph node involvement or suspicion of invasion of the aerodigestive tract or mediastinum. PET/CT is not recommended routinely (R8). Molecular diagnostic studies may be considered in surgical planning when available or performed, but routine use is not recommended (R10).

For T1a tumors, greater emphasis is placed on active surveillance, which yields oncologic and quality of life outcomes comparable with immediate surgery (R11-14). Ultrasound should be performed every 6 months for 1–2 years, then annually by an experienced sonographer. Surgery is recommended if there is evidence of progression or per patient preference. The use of percutaneous ablation of T1a disease as an alternative method to surgery, in select patients, after discussion of risks and benefits, is new (R11).

The role of lobectomy has been expanded, given that recurrence rates are comparable with thyroidectomy, but with fewer complications and less need for thyroid hormone supplementation (R15). Tumors <2 cm without extrathyroidal extension (ETE) or suspicious lymph nodes should be managed with lobectomy unless contraindicated (e.g., contralateral nodules). Lobectomy may also be appropriate for 2–4 cm tumors, if there are no contralateral nodules or ETE. For tumors >4 cm, or those with ETE or nodal disease, total thyroidectomy with lymph node dissection remains recommended.

The 2025 guidelines recognize that central neck dissection (CLND) increases the risk of complications. Prophylactic CLND (pCLND) for cN0 disease does not improve long-term outcomes in T1–T2 tumors and is now more strongly discouraged (R19). For larger cN0 tumors, pCLND may be considered, though evidence is limited and risks must be weighed. Importantly, ipsilateral CLND is recommended when lateral neck nodal disease is present (R20). Therapeutic CLND should be performed if nodal involvement is suspected preoperatively. Abnormal central nodes (cN1a) on imaging warrant CLND to achieve maximal resection.

Intraoperative nerve monitoring of the recurrent laryngeal nerve (RLN) and external branch of the superior laryngeal nerve may be performed during surgery to improve postoperative voice outcomes (R23). The 2025 guidelines also newly advise considering RLN status when deciding on completion thyroidectomy (R16).

New postoperative recommendations include calcium and vitamin D supplementation or selective use based on rapid postoperative parathyroid hormone testing following total or completion thyroidectomy (R24), avoidance of routine surgical drains (R25), and early referral for RLN injury to a speech language pathologist or voice-specialized physician (R26).

Genetic considerations have also been expanded. While the 2015 guidelines noted syndromic DTC, the 2025 update conditionally recommends careful history and physical exam, while ultrasound screening may be performed for familial nonmedullary thyroid cancer when three or more first or second-degree family members are affected (R2-5).

Initial postoperative management of DTC

Updates in histopathologic evaluation reflect the latest World Health Organization classification of thyroid tumors to ensure accurate reporting (R27).[3] Key changes include: renaming encapsulated follicular variant of papillary thyroid cancer (IEFVPTC) without invasion as noninvasive follicular thyroid neoplasm with papillary-like nuclear features, now classified as a low-risk neoplasm not requiring completion surgery or RAI; designating IEFVPTC with minimal capsular invasion as a "favorable outcome" variant; and identifying oncocytic thyroid carcinoma (OTC) and high-grade follicular cell—derived nonanaplastic thyroid carcinoma as distinct subtypes associated with unfavorable outcomes.

The 2025 ATA Risk Stratification System incorporates updated American Joint Committee on Cancer staging, results of molecular profiling if available or performed, and postoperative imaging and serum Tg and TgAb testing, when appropriate, to estimate risk of persistence/recurrence and survival (R28).[4] The 2025 guidelines maintain the same ATA Response Criteria categories—excellent, indeterminate, biochemically incomplete, and structurally incomplete (R29). There are revised Tg thresholds and clearer guidance on applying response criteria to patients following hemithyroidectomy or thyroidectomy without radioactive iodine (RAI).

An updated postoperative risk of recurrence (ROR) stratification system is provided, incorporating additional features found to influence overall risk and refining others (Fig. 2 of 2025 ATA Guidelines). The 3-tiered system from 2015 is now a 4-tier system including low, low-intermediate, intermediate-high, and high ROR categories. Additionally, separate ROR descriptions are provided for papillary thyroid cancer, follicular thyroid cancer (FTC)/IEFVPTC, and OTC. Notable updates include distinguishing N1a as lower risk than N1b disease, assigning a higher ROR for multifocal disease (even for T1 tumors), reclassifying microscopic ETE as a lower-risk feature than previously recognized, and differentiating microscopically positive posterior margins as higher risk than anterior. Notably, BRAF and TERT mutations have been removed from the pictorial depiction of estimated structural recurrence risk. The 2025 guidelines include a formal recommendation on the utility of molecular profiling to further refine recurrence risk if available. Limited vascular invasion (<4 vessels) in OTC and FTC/IEFVPTC is now categorized as low-intermediate risk (10%–15% ROR), whereas in 2015 this was described as a low-risk feature (2%–3% ROR). Additionally, the high-risk category is expanded to include poorly differentiated thyroid cancer (PDTC), high-grade cancers, and extranodal extension.

When estimating ROR, the 2025 guidelines recommend measuring Tg and TgAb 6–12 weeks after surgery rather than 3–4 weeks, reflecting updated data on postoperative Tg nadir kinetics (R30). Additionally, new Tg cutoff values are

Table 1. Notable Changes in Management of DTC

Topic	2015 ATA recommendations	2025 ATA recommendations
NIFTP (R1)	None	New pathological distinction
Germline genetic testing (R2–5)	Discussed without specific recommendations	Germline genetic testing may be of benefit in syndromic familial non-medullary cases. Screening for DICER1 in some instances. Consider neck ultrasound for those with >3 first or second degree family members with DTC.
High volume surgeon (R6)	Discussed in text only	Thyroid cancer cases should be referred to high-volume surgeons (>25–50 cases/year).
Initial preoperative management (R7–10)	Pre-op neck ultrasound to evaluate LNs with biopsy and possible Tg washout. Cross-sectional imaging with CT or MRI for clinical suspicion of advanced disease. Routine Tg not recommended.	Preoperative imaging should also assess for presence of gross extra-thyroidal extension and aero-digestive extent. GPS for chest and abdominal imaging if would affect the extent of surgery.
Active surveillance (R11–14)	Microcarcinomas and patients with high surgical risk can be managed with active surveillance	Added percutaneous ablation as an alternative in select patients. Shared decision-making with patients is essential. For those undergoing AS, monitoring with ultrasound advised and measurement of Tg/TgAb is recommended. Surgery if signs of progression.
Surgical management (R15–16)	Lobectomy preferred for tumors <1 cm. For tumors 1–4 cm either lobectomy or total thyroidectomy	Updated to lobectomy preferred for tumors <2 cm and for 2–4 cm, either lobe or total may be warranted. RLN status should be considered in decision of completion thyroidectomy.
Thyroglossal duct associated cancer (R17–18)	None	Sistrunk procedure recommended along with thyroidectomy if large tumor, evidence of extension outside duct or clinical node involvement.
Lymph node management (R19–20)	Surgery without proph CND is appropriate for T1/T2 tumors with no clinical node involvement. CND should be considered in N1b cases.	Proph CND should not be performed for clinical node-negative T1/T2 tumors. Ipsilateral proph CND is recommended if N1b disease.
Voice/Nerve monitoring (R21–23)	Visualization of RLN should be performed. Intraoperative RLN nerve monitoring may be considered.	In addition to visual ID, RLN monitoring and visual ID and monitoring of external branch of SLN may be performed. Intraoperative vagal or RLN stimulation should be performed prior to removing contralateral lobe.
Parathyroid/Calcium (R24)	Parathyroids should be seen and preserved	In addition, parathyroids removed should be reimplanted. Start calcium/vit D if total or completion thyroidectomy.
Surgical drain (R25)	None	Surgical drains not routinely recommended.
Postoperative voice (R26)	Postoperative voice should be assessed. Laryngeal exam if abnormal	Addition of GPS for timely referral to voice specialist if known RLN injury.
Histopathology (R27)	Tumor classification based on 2015 WHO; NIFTP considered malignant (encapsulated FVPTC) and OTC entitled HCC and a subtype of FTC	WHO updates incorporated: NIFTP reclassified as non-malignant; IEFVPTC with minimal invasion recognized and designated as having "favorable outcome"; OTC and High-grade follicular-derived non-anaplastic thyroid carcinoma recognized and classified as having "unfavorable outcome."

(continued)

Table 1. (Continued)

Topic	2015 ATA recommendations	2025 ATA recommendations
Risk Stratification and response (R28–R30)	3-tier ROR system; Tg/TgAb at 3–4 weeks post-op; Tg cutoffs mainly for thyroidectomy + RAI	4-tier ROR system with refined features and inclusion of new; integrates AJCC 8th ed. and WHO updates; Tg/TgAb at 6–12 weeks; Tg cutoffs now also for total thyroidectomy without RAI (<2.5 ng/mL).
Imaging and FNA immediately following initial surgery (R31)	Limited discussion	More specific guidance: US/cross-sectional imaging before RAI if not "excellent response" or +TgAb; US preferred; cross-sectional imaging (lung/bone) if US inconclusive; PET/CT again recommended for OTC/PDTC with persistent disease concern
Radioactive iodine (R32–43)	Role tied to ROR categories; rhTSH alternative to withdrawal; pregnancy avoidance 6–12 months	More individualized RAI use (esp. low-intermediate / intermediate-high risk); nuanced dosing by ROR tier; OTC—limited outcomes data for RAI; rhTSH now preferred even in high-risk; expanded counseling (SPMs, salivary/lacrimal side effects); pregnancy avoidance ≥6 months; reassure female patients on fertility.
Radiotherapy +/– chemotherapy (R44)	Recommended against for initial postoperative management	EBRT recommended post-surgery for select high-risk patients where progression unlikely salvageable; EBRT/chemo for gross residual or unresectable locally advanced disease.
TSH suppression (R45-46)	Numeric TSH targets: "Low normal" 0.5–2.0 mU/L, 0.1–0.5 mU/L, and <0.1 mU/L.	Binary TSH targets: "within normal reference range" and "below normal reference range."
Serum Tg (R47)	Tg assays should be calibrated; Tg should be monitored at a frequency commiserate with risk	Additions: Trending Tg after lobectomy not routinely recommended; TgAb trends may be helpful, but imaging surveillance should be primary.
De-escalation/discontinuation of monitoring (R48)	None	Stop US surveillance after 5–8 years, and Tg monitoring after 10–15 years (due to "complete response") is reasonable in low-risk DTC s/p total thyroidectomy with an excellent response; monitor US q1-3 years for 5 years after lobectomy if no nodules, or use nodule guidelines if nodules are present.
Role of iodine WBS and PET/CT (R49, R50)	Diagnostic WBS can be done 6–12 months after RAI therapy, SPECT/CT is preferred over planar imaging for RAI localization	Diagnostic WBS does not need to be routinely done, and is not recommended after lobectomy or thyroidectomy without RAI; SPECT-CT + planar imaging can help localize RAI uptake.
Ongoing risk stratification (R51)	Ongoing risk-stratification (response to therapy) is mentioned in main text	Ongoing risk-stratification (response to therapy) is an explicitly recommended strategy.
Rx of locoregional residual, recurrent, or progressive DTC (R52–54)	Re-operation thresholds for cervical nodes: Central neck ≥8 mm or lateral neck ≥10 mm (short axis); RAI and EBRT mentioned in main text	Re-operation thresholds should include size, location, pace, patient factors and preferences, and big picture goals; RAI and EBRT given their own recommendations for use
RAI for locoregional and/or distant mets (R55–57)	Avoid >5.5 GBq/150 mCi if age >70 years; insufficient data to recommend dosimetry; insufficient evidence for rhTSH—consider in select populations	Avoid >5.5 GBq/150 mC if age >70 years or renal insufficiency; use dosimetry if needed, consider its use in other cases; rhTSH or LT4 withdrawal are appropriate.
Empirical RAI for Tg positive, imaging negative DTC (R58)	TSH suppression and surveillance for indeterminate response; empirical RAI if Tg high or rising; okay to repeat if persistent but responding	No significant changes
RAI refractory DTC (R59)	RAIR-DTC is classified via 4 criteria; avoid further I-131 if present	Do not diagnose RAIR-DTC unless status post at least an ablative dose of I-131, consider I-131 response as a spectrum and use criteria to gauge likelihood of response.

(continued)

Table 1. (Continued)

Topic	2015 ATA recommendations	2025 ATA recommendations
General principles of management of RAIR DTC (R60–61)	*BRAF or other mutational testing is not routinely recommended for prognostic purpose*	*Molecular profiling not routinely recommended in absence of planned systemic therapy, but tissue-based biomarker testing should be performed prior to initiating systemic therapy.*
Initial Treatment of RAIR DTC (R62–71)	*MKI therapy should be considered for progressive RAIR DTC*	*MKI therapy is strongly recommended for progressive RAIR DTC without biomarker-linked FDA-approved therapy and lenvatinib is preferred agent.* *The starting dose of 24 mg for Lenvatinib is recommended for most patients and they should be monitored in a specific interval.* *Cabozantinib as 2nd line after lenvatinib in absence of specific biomarker.* *Specific targeted therapies for NTRK, RET, ALK and BRAF V600E should be used for patients with corresponding biomarkers.*
Management of progressive RAIR DTC after initial therapy (R72–77)	*Molecular testing is limited for research purposes* *Redifferentiation therapy needs further research*	*Molecular testing is recommended for progressive disease to identify mechanisms of resistance.* *Redifferentiation therapy for select patients.* *Immune checkpoint inhibitor is recommended for TMB-high or MMR deficient RAIR DTC.* *Recommendation for local therapy for solitary or oligometastatic disease.*
Special considerations (R78-80)	*Bone-modifying agents are recommended for patients with bone metastases*	*Highlights careful modification of dosing schedule for patients at risk for osteonecrosis of jaw*
Management of Pregnant patients with DTC (R81)	*Management of pregnant patients with DTC was incorporated with approaches to patients with malignant or indeterminant cytology from FNA*	*A new recommendation that applies to pregnant patients with DTC—states surgery can be delayed until delivery in most patients.* *US monitoring at least once in early 2nd trimester.* *TSH goals, monitoring with US and Tg for incomplete response should remain the same as determined preconception.*

AJCC, American Joint Committee on Cancer; AS, active surveillance; ATA, American Thyroid Association; CND, central neck dissection; DTC, differentiated thyroid cancer; EBRT, external beam radiotherapy; FDA, Food and Drug Administration; FNA, fine needle aspiration; FTC, follicular thyroid cancer; FVPTC, follicular variant of papillary thyroid cancer; GPS, good practice statement; HCC, hurthle cell carcinoma; IEFVPTC, invasive encapsulated follicular variant of papillary thyroid carcinoma; LN, lymph node; LT4, levothyroxine; MKI, multikinase inhibitor; MMR, mismatch repair; NIFTP, noninvasive follicular thyroid neoplasm with papillary-like nuclear features; OTC, oncocytic thyroid carcinoma; proph, prophylactic; PDTC, poorly differentiated thyroid cancer; R, Recommendation; RAIR, radioactive iodine resistant; RAI, radioactive iodine; RLN, recurrent laryngeal nerve; rhTSH, recombinant human thyrotropin; ROR, risk of recurrence; SLN, superior laryngeal nerve; SPM, secondary primary malignancy; TgAb, antithyroglobulin antibody; TSH, thyrotropin; TMB, tumor mutational burden; Tg, thyroglobulin; US, ultrasound; WHO, World Health Organization; WBS, whole body scan.

recommended for patients after total thyroidectomy who do not receive RAI; < 2.5 ng/mL defines an "excellent" response, 2.5–5 ng/mL defines an "indeterminate" response, and >5 ng/mL defines a "biochemically incomplete" response (Table 9 of 2025 ATA Guidelines).

More specific recommendations for postoperative imaging before radioactive iodine (RAI) are provided (R31). PET/CT is recommended for patients with OTC or PDTC when there is concern for persistent disease, typically indicated by elevated serum Tg and/or TgAb and negative or inconclusive ultrasound. Ultrasound before RAI is recommended for patients who do not have an "excellent" response or have elevated TgAb. Lung and bone cross-sectional imaging is recommended if ultrasound is inconclusive.

The role of RAI remains largely unchanged, but greater flexibility is allowed for individualized decisions in patients with "low-intermediate" and "intermediate-high" risk disease, with increased emphasis on pretreatment counseling. Other notable changes include more nuanced recommendations for each ROR tier (Table 1), a new recommendation highlighting limited outcomes data for use of RAI in OTC and preference for use of recombinant human thyrotropin (rhTSH) over thyroid hormone withdrawal. More attention is also paid to counseling on radiation safety and RAI risks, including second primary malignancy and salivary gland and lacrimal duct complications. Conception avoidance is now recommended for ≥6 months (rather than 6–12) and female patients can be counseled that RAI does not impact fertility at typical administered doses.

The 2025 guidelines permit selective use of external beam radiotherapy (EBRT) after initial surgery, particularly for patients with high-risk features for locoregional progression, especially if not expected to be amenable to salvage surgery. EBRT can now be considered for patients with gross residual disease or locally advanced, unresectable disease where improved locoregional control is anticipated with treatment. The option of concurrent chemotherapy is also newly recommended. Counseling regarding potential toxicity and impact on survival of these interventions is emphasized (R44).

Long-term management of DTC

Several changes are recommended for the long-term management of DTC patients, including new TSH targets (R45), de-escalation of surveillance in low-risk patients (R48), introduction of the term "complete remission "(R48), and recommendations for appropriate use of percutaneous ethanol and radiofrequency ablation (RFA) for recurrences (R52).

The approach to TSH suppression has been simplified into a binary system. Table 9 of the 2025 ATA Guidelines summarizes TSH targets based on the ATA Response Criteria during ongoing reassessment. TSH goals are stratified to "within normal reference range" versus "below normal reference range" due to inconclusive or conflicting data on specific numeric targets. The TSH target after total thyroidectomy with or without RAI is now "within normal reference range" for an excellent or indeterminate response and "below normal reference range" for a biochemically or structurally incomplete response. There are only two response criteria after hemithyroidectomy: excellent and structurally incomplete response. Maintaining TSH suppression for up to 5 years in high-risk patients with excellent or indeterminate response to therapy is no longer explicitly recommended.

Recommendations for Tg measurement are similar to 2015, with the addition of a specific recommendation regarding the limitations of Tg monitoring after hemithyroidectomy; a one-time measurement is recommended as potentially helpful to ensure it is not unexpectedly elevated, although a specific cut-off is not given. Trending TgAb as a surrogate tumor marker is discussed, although imaging is still recommended as the primary surveillance modality if TgAb levels are positive (R47).

Recognizing the long-term burden of surveillance, de-escalation over time is newly given its own recommendation (R48 and Table 11 of the 2025 ATA Guidelines). Patients with a low risk of recurrence and a persistent "excellent" response after total thyroidectomy, with or without RAI (negative imaging, Tg <0.2 ng/mL after RAI or <2.5 ng/mL without RAI), can discontinue ultrasound surveillance after 5–8 years in favor of every 1–2-year Tg monitoring. After 10–15 years of ongoing "excellent" response, patients are newly considered to have entered "complete remission," and biochemical monitoring can also be discontinued. Offering the option to discontinue both biochemical and radiographical surveillance is a major paradigm shift, bringing management of low-risk DTC into closer parity with the larger field of oncology. For patients treated with lobectomy, ultrasound every 1–3 years for 5–8 years is appropriate if the initial study is negative; serial Tg monitoring is not recommended. If nodules remain in the residual lobe, follow-up per ATA nodule guidelines is advised.

Recommendations for use of diagnostic whole-body RAI scan (WBS) have been revised (R49): diagnostic WBS is not recommended after thyroidectomy without RAI, or after hemithyroidectomy. In patients with intermediate-high and high-risk of recurrence, new guidelines recommend that WBS can be done if there is clinical suspicion for recurrence, rather than routinely 6–12 months after RAI. SPECT-CT in addition to (rather than instead of) planar imaging is preferred for anatomical localization of RAI uptake.

Strong emphasis is placed on performing ongoing risk stratification (response to therapy assessment) during long-term disease surveillance, which is an iterative process that informs surveillance intervals and therapeutic management decisions (R51 and Fig. 6 of the 2025 ATA Guidelines).

Size is no longer considered the primary determinant for surgical treatment of recurrent/residual neck nodal disease, in favor of integration of multiple factors including location, pace of growth, and patient factors/preference. Percutaneous ethanol ablation and RFA are now explicitly mentioned as alternatives for managing recurrent/residual DTC in patients at high-risk for complications from reoperation (R52). RAI may be considered for isolated cervical lymph node metastases after local therapy or if local therapy is contraindicated; EBRT may be considered for unresectable or extranodal/soft-tissue invasive locoregional disease (R53).

RAI remains an important treatment modality for recurrent or residual locoregional/distant disease. The 2025 guidelines emphasize care with high (>5.5 GBq/>150 mCi) doses of RAI in elderly patients and those with renal insufficiency, along with use of dosimetry to maximize efficacy and safety if high doses are given (R55). RhTSH is now considered

equally acceptable to levothyroxine withdrawal for I-131 preparation for patients with distant metastases (R55).

The designation of RAI-resistant (RAIR) DTC now requires prior treatment with at least one ablative dose of RAI (R59). Given controversies around the classification of RAIR-DTC and determination of what constitutes a clinically meaningful response, the 2025 guidelines are less definitive, highlighting that criteria suggesting refractory disease should be used to gauge likelihood of responsiveness across a spectrum, rather than serving as definitive evidence of treatment futility.

RAIR DTC and use of systemic therapy

For asymptomatic, stable, or minimally progressive RAIR metastatic DTC, disease monitoring with TSH-suppressive therapy remains an option (R60). Molecular testing is not routinely recommended (R60) but should be performed in progressive cases before systemic therapy (R61). Incorporation of molecular testing to determine initial management strategy for RAIR metastatic DTC reflects advances and FDA approvals of targeted therapies on actionable driver alterations such as NTRK, RET, ALK, and BRAF V600E (R62–71). The guidelines also give a specific recommendation for first line lenvatinib at a starting dose of 24 mg per day, in most patients, and second line cabozantinib for RAIR DTC without actionable alterations (R62–66, R71). For patients with NTRK, RET, or ALK alterations, specific targeted therapy for each is recommended as first line treatment, whereas BRAF V600E targeted therapy is recommended as second line, although it can be considered first line (R67–70). After progression on initial therapy, tissue sampling is recommended for molecular testing to clarify resistance (R72). For patients with high tumor mutational burden, mismatch repair-deficient disease, or tumors with high microsatellite instability, immune checkpoint inhibitor (ICI) therapy is recommended, reflecting tissue agnostic approval of ICI (R73). Redifferentiation therapy for MAPK pathway inhibitor was added as a consideration (R74). Finally, cytotoxic chemotherapy remains an option for RAIR DTC in patients with rapidly progressive and symptomatic disease, though participation in a clinical trial is preferred in this setting (R75). For solitary or oligometastatic (2–5 lesions) disease, focal ablative therapy can be considered (R76). For patients with site-specific symptoms, local therapy can be considered to alleviate symptoms (R77). Bone-modifying agents are still recommended for patients with multiple bone metastases (R78). For brain metastases, surgery and/or stereotactic body radiotherapy remain the mainstay of therapy (R79).

Survivorship

The 2025 guidelines newly recommend discussing potential long-term treatment effects—including voice changes, hypoparathyroidism, scar appearance, neck tightness, dysphagia, psychosocial distress, fatigue, financial impact, and cancer-related worry—with patients (R82). These issues should be incorporated into shared decision-making and supported by access to high-quality resources such as support groups, guidance on managing distress, and local support services (R83–84).

Conclusion

The 2025 ATA guidelines represent a major evolution in the management of DTC, incorporating contemporary evidence, refined pathology classification, and a more personalized, patient-centered approach. These updates are designed to support busy clinicians by providing concise, actionable recommendations that balance oncologic outcomes with quality of life and long-term survivorship.

Authors' Contributions

S.S.P.: Conceptualization, writing—original draft, and review and editing. B.G.: Writing—original draft and review and editing. A.T.: Writing—original draft and review and editing. H.S.: Writing—original draft and review and editing. D.M.: Conceptualization, writing—original draft and review and editing.

Author Disclosure Statement

No competing financial interests exist.

Funding Information

All authors received no funding for this work and declare that neither they nor their institutions received payment or support of any kind for any aspect of this work (including grants, data monitoring board, study design, article preparation, and statistical analysis).

References

1. Haugen BR, Alexander EK, Bible KC, et al. 2015 American Thyroid Association management guidelines for adult patients with thyroid nodules and differentiated thyroid cancer. Thyroid 2016;26(1):1–133.
2. Ringel MD, Sosa JA, Baloch X, et al. 2025 American Thyroid Association management guidelines for adult patients with differentiated thyroid cancer. Thyroid 2025;35(8):841–985.
3. WHO Classification of Tumours Editorial Board. Endocrine and Neuroendocrine Tumours, 5th ed. International Agency for Research on Cancer: Lyon, France; 2025. (WHO classification of tumours series, vol. 10).
4. Tuttle M, Haugen B, Shah J, et al. Thyroid-differentiated and anaplastic carcinoma. In: *AJCC Cancer Staging Manual*, 8th ed. (Amin MB, Greene F, Byrd DR, eds.) Springer International: New York, NY; 2017.

Address correspondence to:
Stephanie Smooke Praw, MD
Division of Endocrinology, Diabetes, and Metabolism
Department of Medicine
University of California Los Angeles
David Geffen School of Medicine
10833 Le Conte Avenue
CHS 57-145
Los Angeles, CA 90095
USA

E-mail: ssmooke@mednet.ucla.edu